CTCFNet: CNN-Transformer Complementary and Fusion Network for High-Resolution Remote Sensing Image Semantic Segmentation

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***Abstract*— Semantic segmentation of high-resolution remote sensing images poses challenges such as scale variability, diverse objects, and obstruction by surface elements. These factors often lead existing methods to suffer from issues like missed and false detections, as well as coarse segmentation boundaries. To tackle these challenges, this article proposes a CNN-transformer complementary and fusion network, termed as CTCFNet. It aims to enhance segmentation accuracy and robustness by extracting and integrating the complementary global and local information from high-resolution remote sensing images. The CTCFNet operates through two primary stages: feature extraction and fusion. In the feature extraction stage, a feature extractor employs convolutional neural network (CNN) and pyramid vision transformer (PVT) blocks to extract both local and global features. A boundary loss is also proposed to improve the segmentation performance for object textures and boundaries. In the feature fusion stage, a feature aggregation module (FAM) is first designed to effectively fuse local and global features at the same scale, facilitating the feature extractor to obtain more comprehensive representations. On this basis, a bi-directional decoder (BiDecoder) reconstructs multiscale features through both top-down and bottom-up directions, resulting in more precise segmentation outputs. Experiments on several high-resolution remote sensing image datasets demonstrate that the proposed method outperforms the state-of-the-art methods in terms of segmentation accuracy and generalization. The code is available at https://github.com/ChenLu0000/CTCFNet.**

***Index Terms*— Complementary information, convolutional neural network (CNN) transformer, feature fusion, semantic segmentation.**

I. Introduction

SEMANTIC segmentation of high-resolution remote sensing images plays a crucial role in remote sensing image analysis, particularly within geographic information systems (GISs), environmental monitoring, urban planning, and agricultural management [1]. By segmenting these images, distinct regions associated with different surface features, such as roads, buildings, water bodies, vegetation, and more, can be identified. This segmentation facilitates a better understanding and analysis of Earth's surface, providing vital assistance for decision support systems. As technology advances, the amount of high-resolution remote sensing images has gradually increased. The complexity of scenes and objects in these images presents significant challenges. Hence, the need for effective semantic segmentation of high-resolution remote sensing images becomes increasingly critical.

Existing methods for remote sensing image segmentation can generally be categorized into traditional and learning-based methods. Traditional methods typically treat pixels independently [2], [3], [4]. These methods rely on basic mathematical models or thresholding techniques for segmentation, which are straightforward to understand and implement. However, these methods often overlook contextual information, making them vulnerable to noise and local variations. On the other hand, learning-based methods encompass a variety of techniques such as convolutional neural networks (CNNs), generative adversarial networks (GANs), Transformer-based methods, and hybrid-architecture methods. These methods leverage automatic feature extraction and adaptive capabilities, leading to significant improvements in segmentation accuracy and robustness compared to traditional methods.

Although existing learning-based methods perform well in remote sensing image segmentation, they still face challenges with high-resolution remote sensing images. First, the complexity of scenes and interference from surface objects in high-resolution images often lead to missed and false detections. Second, these methods struggle with varying object scales, resulting in rough and inaccurate segmentation boundaries. These issues stem from inadequate extraction and insufficient use of both local and global features. Combining CNNs with Transformers for image segmentation has become an active research area. These methods leverage the strengths of both CNNs and Transformers, enhancing segmentation performance by integrating local and global features. Local features help identify fine details and texture information, leading to more precise object segmentation and improved accuracy. Complementarily, global features provide broader contextual information, aiding in understanding overall scene composition and object relationships, thereby enhancing segmentation robustness. Therefore, comprehensive and effective extraction and fusion of both local and global features are

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essential for achieving high-quality remote sensing image segmentation results. However, current CNN-transformer-based methods do not fully reconstruct features and often overlook the details of segmented targets. While these methods have improved segmentation performance to some extent, the segmentation boundaries remain imprecise, and the finer details are lacking.

To address these issues, this article proposes a CNN-transformer complementary and fusion network for semantic segmentation of high-resolution remote sensing images, termed CTCFNet. High-resolution remote sensing images feature complex surface scenes and various object scales. Considering these characteristics, CTCFNet aims to fully utilize local and global information to improve segmentation accuracy for complex scenes and segmentation robustness for various object scales.

1) For *local and global feature extraction*, we propose a feature extractor consisting of multiscale pyramid vision transformer (PVT) blocks and CNN blocks. CNN blocks extract multiscale local features and spatial hierarchical structures, while PVT blocks capture global long-distance dependencies and features through multiscale attention mechanism. These two types of blocks complement each other for comprehensive feature extraction. Unlike existing CNN-transformer methods, a boundary loss function is designed to act on the shallow layer of the feature extractor. This additional supervision helps the model better focus on the boundary detail features of the segmentation target, improving performance for object textures and boundaries.
2) For *local and global feature fusion*, the feature aggregation module (FAM) fuses global and local features extracted by PVT and CNN blocks. Building on this, we design a bidirectional decoding strategy. Existing CNN-transformer methods typically use a top-down unidirectional decoding structure for feature fusion. However, this decoding strategy is insufficient for complex remote sensing images, leading to imprecise segmentation, especially in the detailed parts of targets. In this work, a bidirectional decoder (BiDecoder) is designed, incorporating both top-down and bottom-up decoders, which decode multiscale complementary information in parallel and reconstruct precise semantic segmentation results. The top-down decoder initially handles high-level, abstract semantic features, gradually refining them to lower levels to acquire more specific object semantic information. Conversely, the bottom-up decoder starts with low-level, detailed, and local features of the image, incrementally integrating the detailed and local features.

The main contributions are as follows.

1) We propose a CTCFNet for the semantic segmentation of high-resolution remote sensing images. CTCFNet effectively integrates the strengths of CNNs and Transformers through a well-designed feature extractor and a BiDecoder, resulting in precise segmentation outcomes.
2) For local and global feature extraction, we design a feature extractor that incorporates both CNN and PVT blocks to, respectively, capture complementary local and global information. Additionally, a boundary loss function is introduced to enhance the extraction of texture and boundary information in remote sensing images, leading to more detailed segmentation boundaries.
3) For local and global feature fusion, the FAM first merges the local and global features at the same scale. Following this, a BiDecoder processes the multiscale features in both top-down and bottom-up directions, enabling the complementary integration of different feature types.

The remainder of this article is organized as follows. Section II reviews related research on semantic segmentation of remote sensing images. Section III provides a detailed introduction to CTCFNet. Section IV presents both quantitative and qualitative comparison results of the proposed method against state-of-the-art methods, along with an ablation study to verify the effectiveness of specific designs. Concluding remarks are provided in Section V.

II. Related Works

A. Traditional Segmentation Methods

Traditional remote sensing image segmentation methods included threshold segmentation, region growing, edge detection, clustering, and texture-based methods. Threshold segmentation methods [2], [5], [6], [7], [8] selected a gray level as the threshold, which could be preset or dynamically calculated according to the characteristics of the image. However, threshold segmentation was sensitive to noise, often misclassifying noisy pixels. Region growing algorithms [3], [9], [10], [11], [12] started from a selected seed point and gradually merged surrounding pixels based on a similarity criterion, forming a collection representing the same object or region. These algorithms were effective when object boundaries were clear and less noisy, but their results heavily depended on the seed point selection. Inappropriate seed points could lead to inaccurate segmentation. Edge detection algorithms [13], [14], [15], [16] identified points in an image with significant brightness changes, typically representing object boundaries. However, their performance decreased significantly in complex scenes with occlusions, illumination variations, or intricate textures. Clustering algorithms [4], [17], [18], [19], [20] grouped pixels in an image based on specific characteristics, clustering similar pixels together. The effectiveness of clustering algorithms often depended on the selection of initial cluster centers, with different initial values potentially leading to varied segmentation results. These methods also struggled with flexibility when dealing with objects of varying sizes and shapes. Texture-based image segmentation algorithms [21], [22], [23], [24], [25] distinguished regions by extracting and analyzing texture features within the image. In high-resolution remote sensing images with complex scenes, different objects may exhibit similar texture features, and the same object may show different textures in different regions. This variability makes texture-based segmentation algorithms more challenging to apply effectively.

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B. CNN-Based Segmentation Methods

CNN-based algorithms have achieved significant success across various fields in recent years. fully convolutional network (FCN) [26] was a pioneering deep learning model designed for pixel-level semantic segmentation. As one of the first deep learning models successfully applied to image segmentation, FCN had a profound impact on subsequent research in this area. pyramid scene parsing network (PSPNet) [27], another deep learning model for semantic segmentation, introduced the pyramid pooling module (PPM) to effectively aggregate contextual information from different regions, thereby enhancing the model's ability to capture global information.

DeepLab models [28], [29], [30], [31] represent a suite of state-of-the-art deep learning models specialized in semantic segmentation. These models have undergone several major iterations, each enhancing performance and efficiency. DeepLabV1 [28] introduced atrous convolution, a technique that expanded the receptive field without increasing the number of parameters or computational effort, thereby capturing a broader range of contextual information. Building on this, DeepLabV2 [29] introduced atrous spatial pyramid pooling (ASPP), which effectively captured object information at various scales using atrous convolution at multiple rates. DeepLabV3 [30] further improved ASPP by incorporating global average pooling to capture even more global contextual information. DeepLabV3+ [31] added a decoder module to refine the segmentation results.

For remote sensing image segmentation, Zhan et al. proposed a modified VGG network to segment snow and clouds [32]. Additionally, a global attention feature fusion residual network (GAFFRNet) method [33] was introduced to segment high-resolution remote sensing images. Dai et al. [34] developed the multiscale location attention network (MSLANet) for segmenting buildings and water bodies. However, these methods were constrained by limited receptive fields, made it challenging to capture long-distance dependencies between pixels. Although some methods alleviated this problem by expanding the receptive field, global attention could not be obtained ultimately.

C. Transformer-Based Segmentation Methods

Transformer-based methods have made significant strides in applications and research within the field of image processing in recent years. Initially, Transformer [35] was designed for natural language processing (NLP). With the introduction of the vision transformer (ViT) [36], Transformer-based methods showed promising capabilities in image processing. ViT was the pioneering model to successfully applied Transformer architecture to image classification tasks. It demonstrated that achieving performance on par with or superior to state-of-the-art methods at the time in image classification was feasible using the standard Transformer structure without relying on convolutional layers.

In the realm of image semantic segmentation, segmentation transformer (SETR) [37] adopted ViT directly as its encoder to learn global features. It employed diverse strategies in the decoder phase to translate long-range dependent features into pixel-level predictions. PVT [38] was another Transformer-based model designed for semantic segmentation and object detection. It addressed the limitations of the standard ViT in handling high-resolution images by introducing a pyramid structure. However, these methods neglected the complementarity fusion of local and global information, resulting in unsatisfactory segmentation results.

D. Hybrid-Architecture Segmentation Methods

Recent studies have explored the integration of CNNs, Transformers, and graph methods for semantic segmentation of remote sensing images. Dual-branch network (DBNet) [39] leveraged CNN and Transformer models in parallel for feature extraction from remote sensing images. ST-UNet [40] incorporated the Swin Transformer [41] into the encoder of UNet to extract global and local features. Existing CNN-transformer methods often employed a top-down unidirectional decoding structure for feature fusion, which was insufficient for complex remote sensing images, leading to imprecise segmentation, especially in detailed regions. Moreover, these methods lacked a well-designed boundary loss function. These limitations collectively resulted in poor edge segmentation capabilities, yielding imprecise and detail-deficient results. GraphGST [42], which combined graph theory in a Transformer, introduced absolute positional encoding and graph structure-aware mechanisms, alongside a cross-view comparison learning strategy to enhance classification accuracy and model representational capabilities. WiCoNet [43], GLNet [44], and TCNet [45] treated the small-scale image to be segmented as local and the large-scale image containing it as global. These methods utilized global information from the larger image to assist in segmenting the smaller image. Despite their improvements in segmentation performance, these approaches often neglected segmentation edges, resulting in coarse boundaries and inadequate details.

III. Methodology

This section elaborates on the proposed methodology. First, we present the overall architecture, which comprises a feature extractor and a BiDecoder. We then delve into the specifics of both the extractor and the decoder. Finally, we outline the various loss functions employed in our approach.

A. Overall Architecture

The framework of the proposed method is depicted in Fig. 1. For a given remote sensing image, CTCFNet initially extracts features at multiple scales. An FAM is employed to effectively integrate global and local features. In contrast to existing CNN-transformer methods that typically utilize a top-down unidirectional decoding structure for feature fusion, which proves inadequate for complex remote sensing images and leads to imprecise segmentation, especially in the detailed parts of targets, CTCFNet introduces a BiDecoder in the decoding stage. This arrangement achieves the mutual complementation of high-level and low-level features. To address

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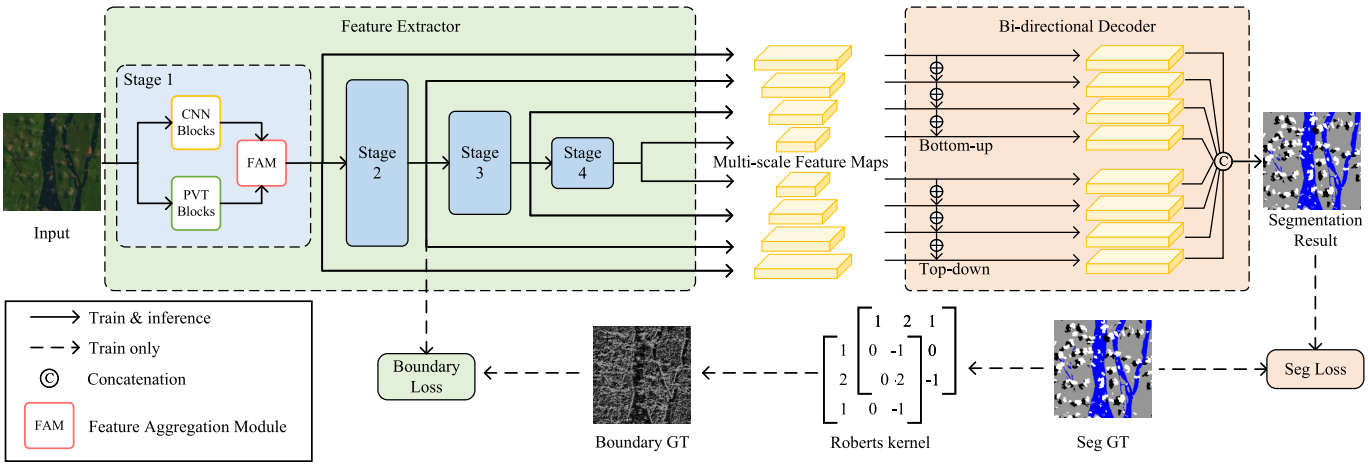


Fig. 1. Overall framework of CTCFNet for semantic segmentation of high-resolution remote sensing images. Multiscale feature maps are composed of the output feature maps of each stage in the feature extractor, and their sizes are 1/4, 1/8, 1/16, and 1/32 of the original image, respectively.

the issue of segmentation boundary roughness observed in existing CNN-transformer methods, a boundary loss function is incorporated to supervise the shallow layer of the feature extractor during training. It helps to extract detailed boundary information from the image, facilitating precise segmentation in regions of interest within high-resolution remote sensing images. Algorithm 1 provides details of the execution process of CTCFNet.

Algorithm 1 Overall Description of CTCFNet

Training phase:

1. Initialize the Feature Extractor and Bi-directional Decoder;
2. Update the parameters of CTCFNet by minimizing $\mathcal{L}$ consisting of $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{boundary}}$ defined in Eq. (13) to obtain the optimal solution.

Testing phase:

1. Feed input $\mathbf{I}$ into the Feature Extractor to generate the multi-scale feature maps ϕ_i;
2. Feed ϕ_i into Bi-directional Decoder to generate the segmentation result $\mathbf{O}$.

B. Feature Extractor

1) PVT and CNN Blocks: The feature extractor integrates PVT blocks and CNN blocks to extract multiscale features of a remote sensing image. Fig. 2(a) and (b) illustrates the specific construction of PVT block and CNN block, respectively. Transformers offer global attention and superior generalization performance, while CNNs provide shift, scale, and distortion invariance, which Transformers lack. Additionally, unlike ViT, PVT incorporates a pyramid structure, making it more suitable for dense prediction tasks. The formulation for a PVT block is as follows:

$$T_1 = \text{Embed}_{\text{pos}}\left[\text{Embed}_{\text{patch}}\left(x_t^i\right)\right] \tag{1}$$

where $x_t^i \in \mathbb{R}^{(H_i W_i)\times C_i}$ denotes the input sequence. $T_1 \in \mathbb{R}^{(H_i W_i)/P_i^2 \times C_i}$ is the output sequence of position embeddings and P_i is the patch size. $\text{Embed}_{\text{patch}}(\cdot)$ denotes patch embeddings. $\text{Embed}_{\text{pos}}(\cdot)$ represents position embeddings. Then, the output sequence of $\text{SRA}(\cdot)$, i.e., $T_2 \in \mathbb{R}^{(H_i W_i)/P_i^2 \times C_i}$ can be defined as

$$T_2 = \text{SRA}[\text{LN}(T_1)] + T_1. \tag{2}$$

The output sequence of the feed forward module, i.e., $x_t^{i+1} \in \mathbb{R}^{(H_{i+1} W_{i+1})\times C_{i+1}}$, can be represented as

$$x_t^{i+1} = \text{FFM}[\text{LN}(T_2)] + T_2 \tag{3}$$

where $\text{LN}(\cdot)$ denotes layer normalization. $\text{FFM}(\cdot)$ is the feed forward module. $\text{SRA}(\cdot)$ is the spatial-reduction attention, which is defined as

$$\text{SRA}(Q, K, V) = \text{Concat}\left(\text{head}_0, \ldots, \text{head}_{N_i}\right) W^O \tag{4}$$

where $Q \in \mathbb{R}^{(H_i W_i)/P_i^2 \times C_i}$, $K \in \mathbb{R}^{(H_i W_i)/P_i^2 \times C_i}$ and $V \in \mathbb{R}^{(H_i W_i)/P_i^2 \times C_i}$ are the input sequences. $\text{Concat}(\cdot)$ represents the concatenation operation. N_i is the number of heads in stage i. Then,

$$\text{head}_j = \text{Attention}\left(QW_j^Q, \text{SR}(K)W_j^K, \text{SR}(V)W_j^V\right) \tag{5}$$

where $W^O \in \mathbb{R}^{C_i \times C_i}$, $W_j^Q \in \mathbb{R}^{C_i \times d_{\text{head}}}$, $W_j^K \in \mathbb{R}^{C_i \times d_{\text{head}}}$, and $W_j^V \in \mathbb{R}^{C_i \times d_{\text{head}}}$ are linear projection parameters. $\text{SR}(\cdot)$ is the operation for reducing the spatial dimension of the input sequence. $\text{Attention}(\cdot)$ is calculated as

$$\text{Attention}(\mathbf{q}, \mathbf{k}, \mathbf{v}) = \text{Softmax}\left(\frac{\mathbf{q}\mathbf{k}^\top}{\sqrt{d_{\text{head}}}}\right)\mathbf{v} \tag{6}$$

where $\boldsymbol{q} \in \mathbb{R}^{(H_i W_i)/P_i^2 \times d_{\text{head}}}$, $\boldsymbol{k} \in \mathbb{R}^{(H_i W_i)/R_i^2 \times d_{\text{head}}}$, and $\boldsymbol{v} \in \mathbb{R}^{(H_i W_i)/R_i^2 \times d_{\text{head}}}$ are the input sequences. R_i is the reduction ratio. d_{head} is the number of heads.

The feature extractor is structured into four stages, each comprising several PVT blocks, CNN blocks, and an FAM. From Stage 1 to Stage 4, the numbers of PVT blocks and CNN blocks are [3, 3, 6, 3] and [2, 2, 4, 2], respectively. The feature maps extracted at these stages are downsampled to 1/4, 1/8, 1/16, and 1/32 of the original image size. The proposed FAM effectively fuses the feature maps of the same size from both PVT blocks and CNN blocks, enhancing feature extraction and integration.

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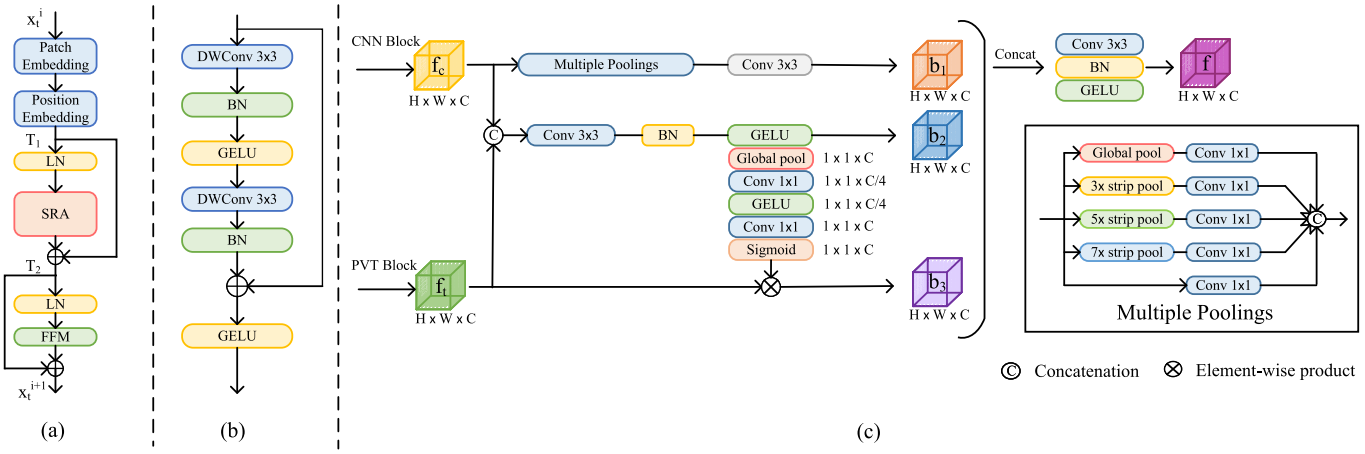


Fig. 2. Illustration of (a) PVT block, (b) CNN block, and (c) FAM. $\text{DWConv}_{3\times3}$ is a depthwise separable convolution with kernel size 3×3; BN represents batch normalization; LN is layer normalization; and GELU is the activation function Gaussian error linear unit.

2) Feature Aggregation Module: Semantic segmentation of high-resolution remote sensing images is a challenging task due to the diversity of object types such as buildings, farmlands, forests, and water bodies. These objects vary significantly in shape, size, texture, and other characteristics, complicating the segmentation process. Existing methods often struggle with missed and false detections when dealing with complex remote sensing images, primarily due to inadequate feature extraction and fusion.

To address these challenges, an FAM is introduced to effectively integrate global and local features. The FAM is designed to combine the strengths of both global and local feature representations, providing a comprehensive feature that enhances the segmentation accuracy. By integrating global features from the PVT blocks and local features from the CNN blocks, the FAM ensures that the model can better differentiate between various objects, even in complex scenarios. This aggregation strategy aims to improve the accuracy and robustness of semantic segmentation in high-resolution remote sensing images, reducing the incidence of missed and false detections.

The FAM comprises three branches, as illustrated in Fig. 2(c). The first branch receives the features from CNN blocks. Multiple poolings are employed to further extract its multiscale local information. This branch mainly includes $3\times$, $5\times$, and $7\times$ strip poolings, as well as a global pooling operation. Strip pooling [46] is a specialized pooling method designed to enhance the representation of spatial information in images. Unlike conventional pooling methods such as max pooling and average pooling, which use fixed-size windows to scan the input feature maps, strip pooling utilizes 1-D strips along specific directions (horizontal or vertical). This technique allows the model to better capture the strip structures in an image that extend horizontally or vertically, such as roads, rivers, and buildings. This capability is particularly important for understanding the spatial structure in high-resolution remote sensing images with complex scenarios. By employing strip pooling of various sizes to capture features at different scales, the model can better adapt to image features of diverse sizes and shapes. After each pooling operation, a 1×1 convolution is applied to adjust the channels of the feature maps. These feature maps are subsequently upsampled to match the input size and concatenated along the channel dimension. The concatenated feature maps are then processed through a 3×3 convolution to produce the output feature maps of the first branch.

The second branch integrates features extracted from both PVT blocks and CNN blocks. Initially, the features from both blocks are concatenated. This concatenated output is then processed through a 3×3 convolutional kernel to generate the branch's output. Transformer-based models excel at capturing global features, while CNN-based models are adept at extracting local features. By combining these two types of features, this branch achieves a more effective fusion of global and local information.

The third branch dynamically adjusts the weights of features extracted by PVT blocks based on the integrated feature map, allowing the module to assign different weights to various feature layers adaptively. Finally, the outputs from all three branches are concatenated along the channel dimension and processed through a convolutional layer to produce the final output of the FAM.

C. Bidirectional Decoder

The design of the decoder is crucial for recovering the details and accuracy of the image. During the encoding phase, the model downsamples the feature maps to extract high-level semantic information, which results in the loss of some spatial detail information. The decoder's goal is to restore these details through upsampling, ensuring that the segmentation results are both semantically accurate and finely detailed at the pixel level. Commonly used decoders follow a top-down structure, starting from high-level features and gradually upsampling and fusing them with low-level features. While this approach provides rich semantic information, it often lacks sufficient detail, leading to rough boundaries in the final segmented objects, especially in high-resolution remote sensing images with complex, variable scenes and significant changes in object scales.

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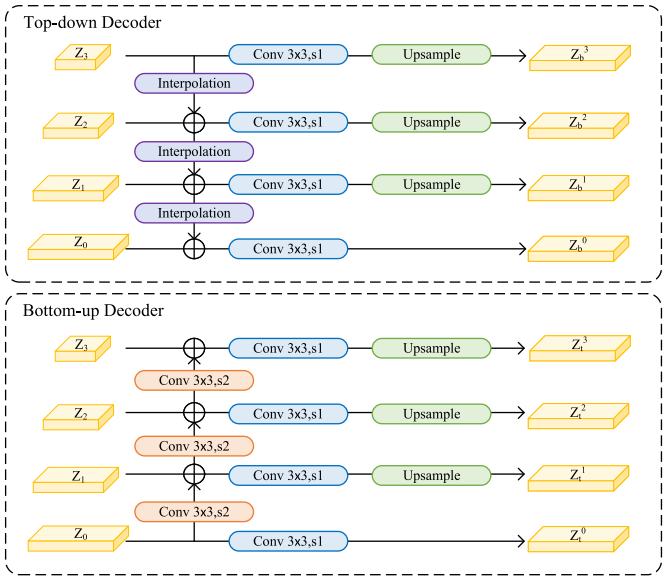


Fig. 3. Illustration of BiDecoder. Interpolation denotes bilinear interpolation and Conv3×3, $s(\cdot)$ represents a convolution with kernel size 3×3 and stride is 1 or 2. Upsample indicates that the current feature map will be enlarged to the same size as z_0.

To address this issue, we propose a BiDecoder, as shown in Fig. 3. The BiDecoder processes multiscale feature maps in both top-down and bottom-up directions, allowing for the fine segmentation of objects while accurately recognizing their categories. This approach ensures a more comprehensive integration of high-level semantic and low-level detailed information, resulting in more precise segmentation outcomes.

Concretely, the BiDecoder consists of a top-down decoder and a bottom-up decoder operating in parallel. For the four multiscale feature maps $[z_0, z_1, z_2, z_3]$ extracted from the feature extractor, the top-down decoder begins with the deepest feature map z_3, which contains the most abstract and semantic-rich information. It then gradually upsamples and combines these with shallower feature maps. This process allows the deeper semantic features to refine the shallower detailed features, aiding the model in understanding the regions of interest and the scene structure in remote sensing images.

Simultaneously, the bottom-up decoder starts with the shallowest feature maps z_0 and gradually downsamples and fuses them with deeper feature maps. This approach focuses on enhancing the details and local features of the feature maps. The detailed features in the shallower layers are progressively passed to the deeper layers, enriching the deeper features with detailed information. This process is crucial for the model to capture fine edge and texture details in the image, facilitating the accurate identification of objects in complex scenes.

D. Loss Function

1) Segmentation Loss: For the loss function, we first define the basic segmentation loss as

$$\mathcal{L}_{\text{seg}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\cdot\log\left(p_{i,c}\right) \tag{7}$$

where $\mathcal{L}_{\text{seg}}$ is the cross-entropy loss. N is the number of pixels. C is the number of classes. $y_{i,c}$ is the ground truth indicating whether the ith pixel belongs to category c. $p_{i,c}$ is the predicted probability, which indicates that the ith pixel belongs to category c.

2) Boundary Loss: Moreover, accurately segmenting object boundaries is a significant challenge in semantic segmentation of high-resolution remote sensing images. Objects such as houses, trees, and vegetation often overlap or are closely arranged in these images. Precise boundary segmentation is crucial for distinguishing between these objects and reducing false detections. Existing methods often struggle with false detections in complex remote sensing scenes, resulting in rough segmentation boundaries, particularly for small objects like buildings, rivers, thin clouds, and similar features.

To cope with this challenge, as shown in Fig. 1, a boundary loss function is added as an additional constraint in Stage 2 of the feature extractor. This method enhances the training process by focusing on the boundary region, enabling more effective weight adjustments. To create the boundary ground truth, we upsample the segmentation ground truth using different strides of the Roberts kernel and aggregate the results. We evaluated several commonly used edge detection operators (Roberts, Laplace, Sobel, etc.) for detecting the model boundary. Based on the experimental analysis, we finally select the Roberts kernel for extracting the segmentation ground-truth boundaries. The process for generating the boundary ground truth is outlined as follows:

$$\text{GT}_b^i = \begin{cases} K_{\text{Roberts},s1}\left(\text{GT}_{\text{seg}}\right), & i=1 \\ \text{Upsample}\left[K_{\text{Roberts,si}}\left(\text{GT}_{\text{seg}}\right)\right], & i=2,4 \end{cases} \tag{8}$$

where GT_{seg} denotes the segmentation ground truth. $K_{\text{Roberts,si}}(\cdot)$ represents the Roberts kernel with a stride of $i \in \{1,2,4\}$. GT_b^i represents the pseudo-label generated by the Roberts kernel with the stride of $i \in \{1,2,4\}$.

Considering the results of different strides, the resultant boundary ground truth GT_b can be expressed as

$$\text{GT}_b = \text{GT}_b^1 + \text{GT}_b^2 + \text{GT}_b^4. \tag{9}$$

Based on the above boundary ground truth, the boundary loss $\mathcal{L}_{\text{boundary}}$ is defined through two aspects

$$\mathcal{L}_{\text{boundary}} = \mathcal{L}_{\text{Dice}} + \alpha\mathcal{L}_{\text{BCE}} \tag{10}$$

where α is the hyperparameter that controls the trade-off between $\mathcal{L}_{\text{BCE}}$ and $\mathcal{L}_{\text{Dice}}$.

$\mathcal{L}_{\text{Dice}}$ is the Dice loss, which is defined as

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\cdot|X\cap Y|}{|X|+|Y|} \tag{11}$$

where X represents the ground truth and Y represents the prediction result. $|X\cap Y|$ denotes the intersection between X and Y, which is the number of pixels that are positive in both. $|X|$ and $|Y|$ represent the size of X and Y, respectively, which is the total number of pixels.

$\mathcal{L}_{\text{BCE}}$ is the binary cross-entropy loss, which is defined as

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i\cdot\log(p_i) + (1-y_i)\cdot\log(1-p_i)\right] \tag{12}$$

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where N is the number of samples. y_i is the ground truth for the ith sample. p_i is the probability predicted by the model that the ith sample belongs to the positive class.

3) Final Loss: The final loss function comprises a weighted sum of the boundary loss function and segmentation loss function

$$\mathcal{L} = \mathcal{L}_{\text{seg}} + \beta \mathcal{L}_{\text{boundary}} \tag{13}$$

where $\mathcal{L}_{\text{boundary}}$ is the proposed boundary loss function. β is the hyperparameter that controls the trade-off between $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{boundary}}$.

IV. Experiments

In this section, we evaluate CTCFNet against several state-of-the-art algorithms using three remote sensing image datasets. We begin by describing the datasets and providing implementation details. Next, we perform both quantitative and qualitative comparisons with existing algorithms. Finally, we verify the effectiveness of specific designs through ablation studies.

A. Datasets

We perform the experiments on three remote sensing image datasets, including L8SPARCS, ISPRS Potsdam, and HRC_WHU. The detailed description and introduction of these methods are summarized in Table I.

1) L8SPARCS Dataset: To assess the segmentation performance on multispectral high-resolution remote sensing images, we conducted experiments using the Landsat8 spatial procedures for automated removal of cloud and shadow (L8SPARCS) dataset. Landsat-8, an Earth observation satellite equipped with the operational land imager (OLI) and thermal infrared sensor (TIRS), provides detailed surface observations. The OLI captures high-resolution remote sensing images, which contain nine spectral bands covering the range from the visible to the near-infrared and short-wave infrared. The specific parameters for these bands are shown in Table II. The spatial resolution of most bands is 30 m, and that of the blue band for water and coastline monitoring and the deep blue band for atmospheric corrections is 15 m. TIRS captures thermal radiation and its spatial resolution is 100 m. This dataset has 80 scene images with a resolution of 1000 $\times$ 1000 pixels and contains a total of 5 semantic categories, including *Snow/Ice, Water, Cloud, Cloud Shadow*, and *Land*. As shown in Fig. 4, we cropped these images into patches with a resolution of 256×256 pixels and applied data augmentation techniques, including horizontal flip, vertical flip, random rotation, etc. We allocated 80% of the patches for training and 20% for validation.

2) ISPRS Potsdam Dataset: In the International Society for Photogrammetry and Remote Sensing 2-D Semantic Labeling Contest-Potsdam (ISPRS Potsdam) dataset, the images encompass most urban areas of Potsdam, Germany, including residential, industrial, and vegetative regions. As detailed in Table I, the dataset consists of six semantic categories: *impervious surface, building, low vegetation, tree, car*, and *clutter*. The original dataset contains 38 IR-R-G images with a spatial resolution of 6000 $\times$ 6000 pixels. We cropped these images into smaller images with a resolution of 256 $\times$ 256 pixels and applied data augmentation techniques (including random flipping, rotation, etc.) This process yielded a total of 17 280 small images, with 13 824 images used for training and 3456 images reserved for validation, maintaining a 4:1 ratio. Fig. 4 displays some of the images and corresponding ground truths from the dataset.

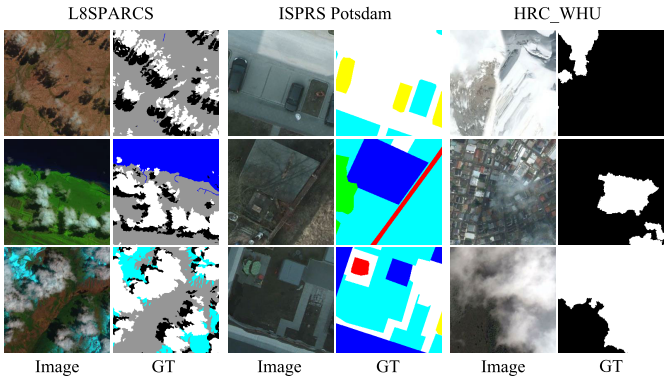


Fig. 4. Samples in the L8SPARCS, ISPRS Potsdam and HRC_WHU datasets.

3) HRC_WHU Dataset: The high-resolution cloud cover validation (HRC_WHU) dataset is sourced from Google Earth and comprises 150 RGB images with a spatial resolution of 1280 $\times$ 720 pixels. These images primarily feature five types of scenes: *Water, Vegetation, Urban, Snow/Ice*, and *Barren*. The dataset is mainly intended for cloud segmentation, containing two semantic categories: Cloud and Background. Remote sensing imagery experts from Wuhan University digitized the relevant reference cloud masks. Fig. 4 presents some images and corresponding ground truths from the dataset. We cropped the images into patches with a resolution of 256 $\times$ 256 pixels and applied randomly flipping and rotate to enhance the model's generalization capabilities.

B. Implementation Details

All experiments are conducted on an NVIDIA RTX 4090 using the PyTorch platform. Adam is employed as the optimizer, and the learning rate is initially set as 0.0001 with the "poly" decay strategy. In addition, to mitigate overfitting, L2 regularization is applied with a weight decay set to 0.01. The batch size is set to 8, and the number of epochs is set to 200. The hyperparameters are configured as $\alpha = 1$, $\beta = 1$.

To quantitatively evaluate the semantic segmentation performance, we use four metrics: accuracy, precision, recall, F1-score, and intersection over union (IoU). The mean values of these metrics are employed for comparison. The definitions of these metrics are as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{14}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F_1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

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TABLE I
CHARACTERISTICS OF REMOTE SENSING IMAGES IN DIFFERENT DATASETS

Dataset	Band	Spatial Resolution	Category	Data Partitioning		
				Train	Val	Test
L8SPARCS	10	1000×1000	Snow/Ice, Water, Cloud, Cloud Shadow, Land	7870	2375	1000
ISPRS Potsdam	3	6000×6000	Impervious Surface, Building, Low vegetation, Tree, Car, Clutter	13824	2419	1037
HRC_WHU	3	1280×720	Cloud, Background	3374	590	354

TABLE II
CHARACTERISTICS OF REMOTE SENSING IMAGES IN L8SPARCS DATASET

Band	Coastal	Blue	Green	Red	NIR	SWIR-1	SWIR-2	Pan	Cirrus	TIRS-1
Wavelength (μm)	0.43–0.45	0.45–0.51	0.53–0.59	0.64–0.67	0.85–0.88	1.57–1.65	2.11–2.29	0.50–0.68	1.36–1.38	10.6–11.19
Resolution (m)	30	30	30	30	30	30	30	15	30	100

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{18}$$

where true positive (TP) is the number of pixels correctly predicted to be in the positive category. True negative (TN) is the number of pixels correctly predicted to be in the negative category. False positive (FP) is the number of pixels incorrectly predicted to be in the positive category. False negative (FN) is the number of pixels incorrectly predicted to be in the negative category.

C. Evaluation on L8SPARCS Dataset

1) Qualitative Comparison: Fig. 5 presents the qualitative comparison results between the proposed method and other state-of-the-art methods.

The proposed method excels at accurately segmenting objects of varying scales. The first four rows of Fig. 5 primarily illustrate the segmentation outcomes for different scales of clouds, cloud shadows, snow/ice, and land. Notably, methods such as BiSeNetV2, DeepLabV3+, GLNet, and MSLANet exhibit numerous missed detections and produce rough object boundaries. Transformer-based methods like PVT, Segformer, and Swin Transformer, which leverage global attention mechanisms, demonstrate improved contextual information capture and reduced missed detections compared to these earlier approaches.

Additionally, methods like GLNet, WiCoNet, and DBNet, which integrate global and local information, perform better than the aforementioned methods. However, these methods still struggle with adequately capturing detailed object boundary information due to insufficient processing and optimization of detailed features.

In contrast, the proposed method effectively utilizes global information from the PVT blocks and detailed local information from the CNN blocks. This enables the segmentation of multiscale objects, significantly reducing the issue of extensive missed detections seen in existing methods. Moreover, the BiDecoder in the proposed method meticulously decodes both deep semantic features and shallow spatial details, enhancing the precision of the final segmentation results. This bidirectional process ensures accurate segmentation of clouds, cloud shadows, snow/ice, and land across various scales, enriching object boundaries with finer details. Consequently, our method achieves more refined segmentation, surpassing the performance of current methods.

The proposed method has the fewest missed and false detections. The fifth and sixth rows of Fig. 5 depict segmentation results for scenarios involving water, clouds, cloud shadows, and land. BiSeNetV2, DeepLabV3+, GLNet, MSLANet, and SegFormer show extensive areas of missed detections due to ineffective integration of semantic and spatial detail information, often misidentifying water as land or vice versa. In contrast, Swin Transformer, PVT, WiCoNet, DBNet, and the proposed method do not exhibit such issues. However, for detailed segmentation of objects, these existing methods fall short compared to the proposed method.

The proposed method excels because it leverages a feature extractor composed of PVT blocks and CNN blocks to capture both global and local features from the image, allowing the model to gather comprehensive information about different aspects of the object. Additionally, the FAM is designed to effectively integrate the global information and spatial details extracted by these blocks, thereby enhancing the model's feature extraction capabilities. This dual approach significantly improves the accuracy and detail of the segmentation results.

CTCFNet excels in segmenting small and narrow objects that are challenging to recognize. The last two rows of Fig. 5 demonstrate the segmentation capabilities of various methods on narrow water and land. CNN-based methods like BiSeNetV2 and MSLANet struggle in these scenarios due to their lack of global information. While DeepLabV3+ uses atrous convolution to expand the receptive field and improve segmentation capabilities, its results are still inadequate.

Transformer-based and hybrid-architecture methods benefit from global features, enabling them to roughly segment out narrow water bodies and land, but they fall short in capturing finer details. The proposed method addresses this issue by introducing an additional boundary loss function during the training phase, enhancing the ability to extract spatial detail information from the shallow layers of the feature extractor. This approach significantly improves the extraction of object

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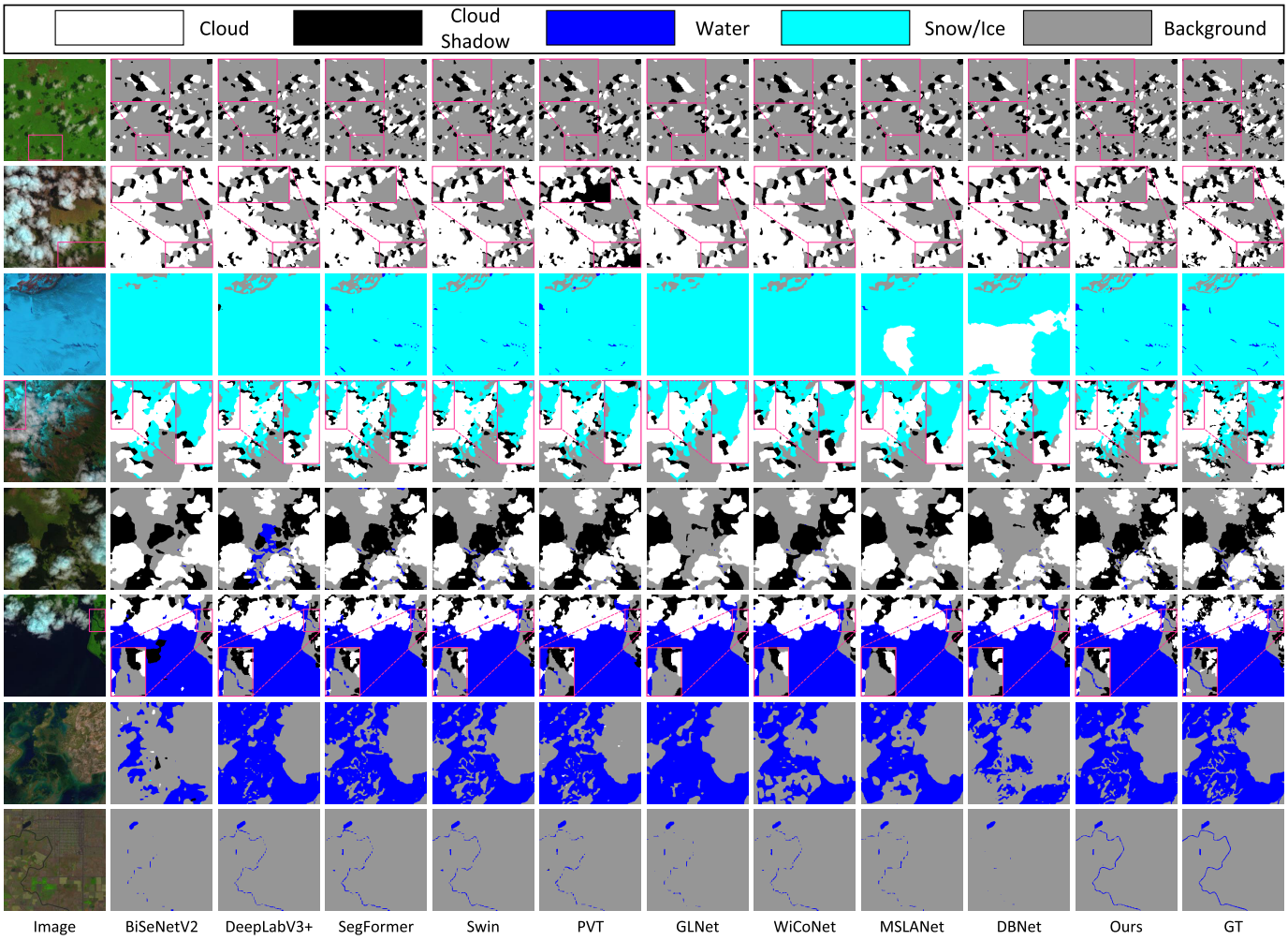


Fig. 5. Qualitative comparisons on the L8SPARCS dataset.

boundary and texture information. As a result, the proposed method achieves more precise segmentation of narrow water bodies and land compared to existing methods.

2) Quantitative Comparison: Table III presents the quantitative comparison results of various method on the L8SPARCS dataset. In terms of class accuracy, the proposed method surpasses other methods, achieving precision rates of 79.6%, 92.03%, and 97.53% for cloud shadow, water, and land, respectively. While the proposed method does not achieve the highest accuracy for cloud and snow/ice, it closely approaches the best-performing methods, with rates of 92.34% and 97.03%, respectively.

From an overall metrics perspective, the mIoU of methods like CCNet, BiSeNetV1, BiSeNetV2, DeepLabV3, FPN, STDC, modified VGG, and GAFFRNet all fall short of 75%. The mIoU for FCN8S, DANet, DeepLabV3+, HRNet, OCRNet, UPerNet, Nonlocal, iSANet, MSLANet, WiCoNet, and GLNet fall between 75% and 85%. These methods enhance the model's representation ability by expanding the receptive field, utilizing multiscale features, and fusing global and local information. Their segmentation results are slightly better than the aforementioned methods. As shown in Fig. 5, the instances of missed and false detections have decreased compared to previous methods but still persist.

In contrast, SegFormer, Swin Transformer, and the proposed method are among the top-performing approaches, with mIoU exceeding 85%. The proposed method achieves the highest scores for mean accuracy (mAcc), mean recall (mRecall), and mean IoU (mIoU), which are 94.78%, 93.39%, and 86.42%, respectively. Although the mPrecision (mean precision) and mF1score (mean F1 score) are not the highest, they are very close to the top-performing methods, reaching 91.70% and 90.69%, respectively.

D. Evaluation on ISPRS Potsdam Dataset

1) Qualitative Comparison: Fig. 6 illustrates the segmentation results of various methods on the ISPRS Potsdam dataset.

The proposed method demonstrates robust segmentation performance across objects of varying scales, as illustrated in the first two rows of Fig. 6. Methods like BiSeNetV2, DeepLabV3+, and SegFormer struggle with effectively segmenting cars, exhibiting numerous missed detections and hollow areas within their segmented regions. This suggests limitations in their multiscale feature extraction capabilities. Moreover, SegFormer, Swin Transformer, and MSLANet face challenges in accurately distinguishing low vegetation from impervious surfaces and in correctly identifying trees, often leading to misclassifications.

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TABLE III
QUANTITATIVE COMPARISONS ON THE L8SPARCS DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Class Accuracy					Overall Metrics				
	Cloud	Cloud Shadow	Snow/Ice	Water	Land	mAcc	mPrecision	mRecall	mF$_1$-score	mIoU
FCN8S [26]	88.46	72.26	92.45	80.68	94.62	90.72	86.31	85.69	86.00	76.23
DANet [47]	90.06	77.22	**97.81**	80.79	97.34	93.44	91.99	88.65	90.15	82.58
CCNet [48]	90.66	64.52	92.44	64.44	96.98	90.98	88.95	81.81	84.69	74.64
BiSeNetV1 [49]	89.19	73.71	51.81	80.37	97.06	89.38	87.54	78.43	81.75	70.35
BiSeNetV2 [50]	84.79	67.53	64.81	74.07	97.41	88.86	87.49	77.72	81.78	69.94
DeepLabV3 [30]	89.85	77.09	72.72	77.58	96.39	90.76	88.52	82.72	85.15	74.72
DeepLabV3+ [31]	92.36	73.18	94.95	84.92	97.19	93.50	92.29	88.52	90.23	82.70
HRNet [51]	92.73	75.98	89.73	85.21	96.48	93.00	91.04	88.03	89.45	81.41
FPN [52]	90.35	76.92	62.96	83.17	97.14	90.90	88.94	82.11	84.86	74.42
PSPNet [27]	88.19	70.30	94.52	75.60	96.83	91.69	90.33	85.09	87.42	78.42
OCRNet [53]	90.66	77.31	95.61	85.58	97.38	93.71	92.32	89.31	90.73	83.49
UPerNet [54]	87.43	78.19	92.07	77.06	95.47	91.24	89.07	86.04	87.23	78.17
NonLocal [55]	86.07	75.52	91.47	74.66	97.16	91.60	89.72	84.97	87.08	77.85
PSANet [56]	93.08	73.49	94.43	80.94	96.77	93.14	92.00	87.74	89.66	81.79
STDC [57]	81.62	63.45	62.45	79.42	97.22	87.97	85.62	76.83	80.59	68.50
ISANet [58]	91.28	77.10	94.57	82.45	97.28	93.48	92.06	88.53	90.18	82.66
SegFormer [59]	90.18	81.32	95.63	86.04	96.84	93.63	91.59	90.00	90.79	83.56
Swin [41]	89.13	74.94	97.00	82.63	97.32	93.12	91.37	88.20	89.63	81.78
modified VGG [32]	85.55	58.53	94.87	79.35	95.98	89.99	82.85	86.38	79.36	74.00
GAFFRNet [33]	86.97	59.00	85.21	78.06	94.47	88.62	80.74	86.70	78.56	72.32
MSLANet [34]	92.07	72.28	79.88	84.24	95.68	91.32	89.16	84.83	86.73	77.04
DBNet [39]	91.04	78.50	96.63	88.53	97.11	93.95	90.36	92.26	88.07	84.40
WiCoNet [43]	**93.09**	78.01	97.55	85.94	96.96	94.14	**92.68**	90.31	**91.40**	84.60
GLNet [44]	92.56	72.31	86.67	72.57	96.42	91.67	89.77	84.11	86.53	76.97
CTCFNet (ours)	92.34	**79.60**	97.03	**92.03**	**97.53**	**94.78**	91.70	**93.39**	90.69	**86.42**

In contrast, our method leverages PVT blocks and CNN blocks to comprehensively extract global and local image features. The inclusion of strip poolings within the FAM further enhances the model's ability to capture multiscale information from local features, thereby improving segmentation across objects of different scales. The second branch of the FAM integrates both global and local features extracted by PVT blocks and CNN blocks, enabling a more holistic understanding of the scene. Meanwhile, the third branch of the FAM adjusts the weights of global features based on the integrated features from the second branch, effectively modulating the contribution of different feature layers. As a result, our proposed method achieves precise segmentation for objects such as cars, trees, and low vegetation, significantly reducing instances of misclassification compared to existing approaches.

CTCFNet demonstrates superior adaptability across various complex scenarios. The third to fifth rows of Fig. 6 primarily present the segmentation results for impervious surfaces, buildings, low vegetation, trees, and clutter. In these results, methods like BiSeNetV2, SegFormer, Swin Transformer, PVT, GAFFRNet, and MSLANet exhibit numerous areas where building, clutter, and impervious surfaces are confused. While GLNet, WiCoNet, and DBNet show enhanced performance due to their integration of global and local features, they still exhibit minor areas of false detections, attributed to incomplete utilization of multiscale features in feature reconstruction.

Feature reconstruction during the decoding phase plays a pivotal role, where our proposed method employs a BiDecoder for this purpose. The top-down decoder gradually upsamples deep semantic features and merges them with shallow features, thereby enriching the spatial detail information within the deep semantic features. Simultaneously, the bottom-up decoder progressively downsamples shallow features and integrates them with deep semantic features to supplement spatial details lacking in the deeper layers. This BiDecoder effectively integrates high-level semantic information with low-level spatial details, facilitating precise categorization of objects. Among these methods, our approach proves to be the most effective in achieving accurate judgments across object categories.

The proposed method excels in boundary segmentation of objects, as illustrated in the last three rows of Fig. 6. These results primarily showcase the boundary segmentation capabilities across various methods for different object categories, including the segmentation performance for small objects. Notably, many methods such as BiSeNetV2, SegFormer, Swin Transformer, PVT, GLNet, WiCoNet, MSLANet, and DBNet struggle with false detections along the boundaries of low vegetation. Moreover, the segmentation results for elongated and circular clutter suffer from inadequate edge details.

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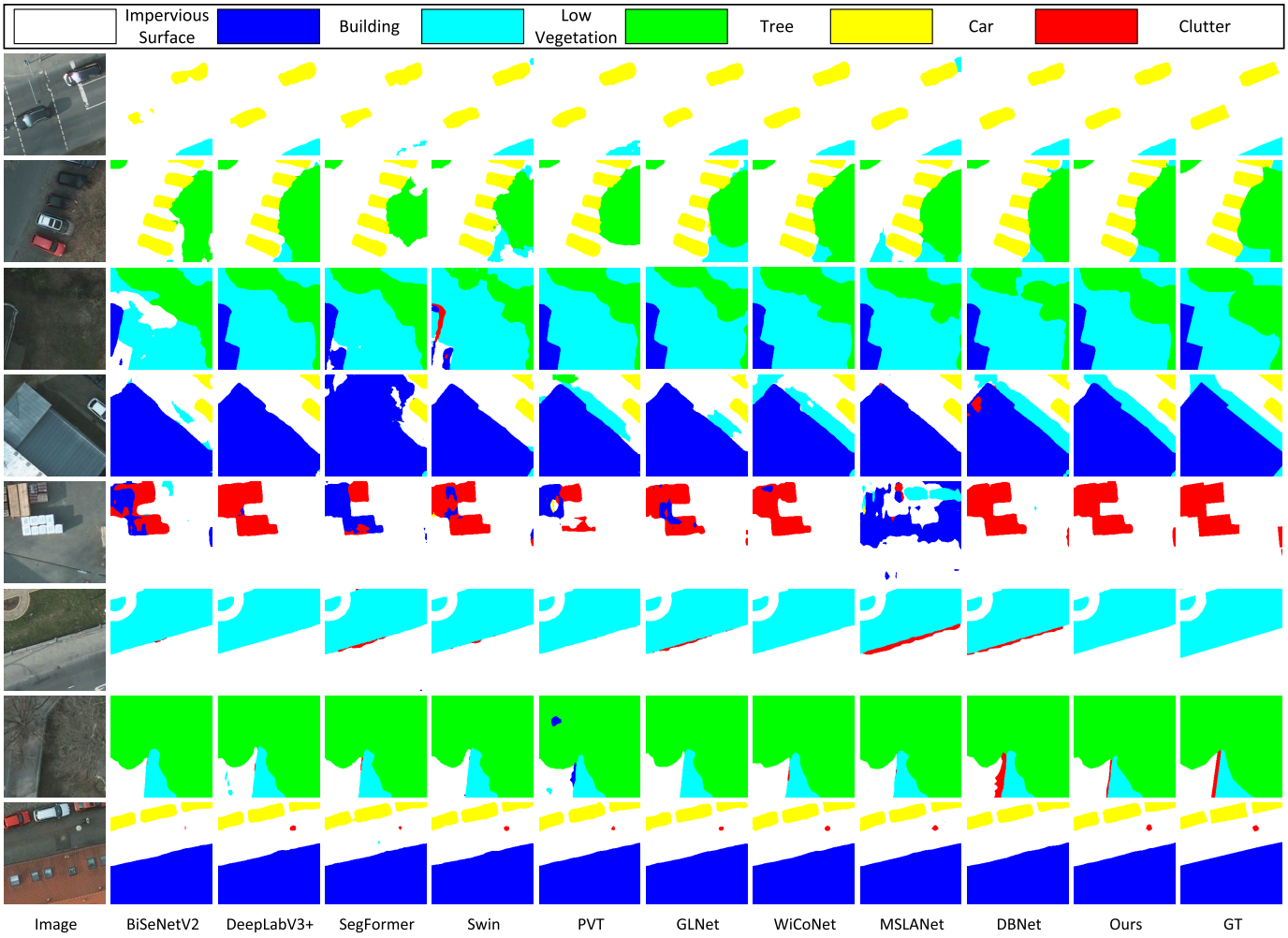


Fig. 6. Qualitative comparisons on the ISPRS Potsdam dataset.

The proposed method excels in boundary segmentation of objects, as illustrated in the last three rows of Fig. 6. These results primarily showcase the boundary segmentation capabilities across various methods for different object categories, including the segmentation performance for small objects. Notably, many methods such as BiSeNetV2, SegFormer, Swin Transformer, PVT, GLNet, WiCoNet, MSLANet, and DBNet struggle with false detections along the boundaries of low vegetation. Moreover, the segmentation results for elongated and circular clutter suffer from inadequate edge details.

To address these challenges, our proposed method integrates a boundary loss function as an additional constraint, significantly enhancing the model's capability to accurately segment object boundaries. This approach improves the model's ability to capture spatial detail information, particularly focusing on boundary details, thereby reducing false detections along object boundaries. The proposed method achieves precise segmentation of object boundaries, especially for small objects.

2) Quantitative Comparison: Table IV presents the quantitative comparison results of these methods. While the proposed method does not achieve the highest accuracy in every category, the differences compared to the top-performing methods are minimal. Notably, methods achieving the highest class accuracy often exhibit significant trade-offs: BiSeNetV1 excels in clutter precision but lags behind in low vegetation segmentation; OCRNet, ISANet, and Swin Transformer each excel in specific categories like car, impervious surface, building, and low vegetation, but struggle with precision in clutter segmentation. Such imbalances may limit these models' practical applicability across diverse inputs.

In contrast, the proposed method demonstrates robust segmentation capabilities across all categories. This is underscored by its leading overall metrics: 79.69% in mAcc, 78.34% in mPrecision, 72.71% in mRecall, 74.97% in mF1score, and 61.90% in mIoU, which are the highest among the evaluated methods. Furthermore, as illustrated in Fig. 6, the proposed method consistently outperforms existing methods by providing more accurate segmentation in each category, offering detailed boundary information, and minimizing instances of false and missed detections.

E. Evaluation on HRC_WHU Dataset

1) Qualitative Comparison: CTCFNet demonstrates exceptional performance in cloud segmentation across various scenarios. Fig. 7 presents the segmentation results of different methods on the HRC_WHU dataset, encompassing typical scenes such as barren, vegetation, snow, urban, and water. It is evident that BiSeNetV2, SegFormer, PVT, and

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TABLE IV
QUANTITATIVE COMPARISONS ON THE ISPRS POTSDAM DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Class Accuracy						Overall Metrics				
	Impervious Surface	Building	Low Vegetation	Tree	Car	Clutter	mAcc	mPrecision	mRecall	mF$_1$-score	mIoU
FCN8S [26]	85.87	85.59	79.11	70.71	77.20	19.32	78.53	77.07	69.63	71.68	58.92
DANet [47]	84.42	88.43	79.06	72.94	77.6	32.90	**79.76**	**79.20**	72.56	74.84	61.69
CCNet [48]	84.63	89.18	77.21	69.77	74.83	24.52	78.64	77.30	70.02	72.42	59.34
BiSeNetV1 [49]	80.14	86.59	59.46	72.93	75.97	**43.77**	74.33	71.13	69.81	69.86	56.16
BiSeNetV2 [50]	85.05	84.15	71.08	71.68	76.04	12.32	76.07	73.59	66.72	68.42	55.48
DeepLabV3 [30]	82.18	88.81	78.97	69.79	79.52	35.56	78.75	75.46	72.47	73.71	60.64
DeepLabV3+ [31]	86.10	86.93	76.06	73.75	74.58	34.65	79.46	77.55	72.01	74.31	61.29
HRNet [51]	81.68	89.40	74.54	70.25	78.46	21.45	77.22	75.84	69.30	71.17	57.98
FPN [52]	85.93	85.77	76.12	68.84	76.37	27.22	78.00	76.24	70.04	72.30	59.08
PSPNet [27]	86.20	85.56	76.14	71.75	78.87	24.12	78.44	76.11	70.44	72.38	59.50
OCRNet [53]	85.49	87.53	77.06	72.69	**80.05**	23.35	79.03	77.51	71.03	72.87	60.00
UPerNet [54]	83.72	82.85	78.37	69.23	78.57	42.01	77.87	75.07	72.46	73.53	59.98
NonLocal [55]	83.84	86.59	73.80	74.14	75.94	33.38	78.32	75.41	71.34	73.09	59.78
PSANet [56]	**88.05**	84.95	70.16	**75.01**	77.32	33.85	78.60	76.55	71.56	73.60	60.30
STDC [57]	77.08	84.91	72.39	67.45	73.15	32.59	74.18	70.39	67.93	68.77	55.04
ISANet [58]	79.39	**89.78**	76.99	70.50	73.18	16.30	76.81	72.60	67.69	69.01	55.79
SegFormer [59]	82.06	90.21	72.23	69.89	76.04	28.39	77.25	74.14	69.80	71.50	57.99
Swin [41]	87.25	82.74	**79.81**	70.49	77.14	22.73	78.55	77.19	70.03	72.12	59.02
modified VGG [32]	71.17	82.06	61.47	59.53	57.55	14.38	66.84	63.10	57.69	59.45	44.40
GAFFRNet [33]	82.95	85.96	78.28	62.78	72.58	3.74	75.36	73.52	64.38	64.96	52.50
MSLANet [34]	77.37	89.29	73.60	66.52	75.99	40.16	75.82	73.57	70.49	71.67	57.66
DBNet [39]	83.74	86.59	79.49	72.73	77.89	30.07	79.04	75.74	71.75	73.37	60.50
WiCoNet [43]	82.55	89.74	75.22	74.92	77.07	30.90	78.92	76.19	71.73	73.49	60.34
GLNet [44]	87.09	84.70	78.53	67.91	73.08	13.40	77.75	76.69	67.45	69.66	57.02
CTCFNet (ours)	87.25	86.69	76.24	72.37	77.68	36.05	79.69	78.34	**72.71**	**74.97**	**61.90**

MSLANet fail to produce satisfactory segmentation results across all scenes, particularly struggling with small-scale and thin clouds. DeepLabV3+, Swin Transformer, GLNet, WiCoNet, and DBNet exhibit slightly improved outcomes but still show coarse segmentation around cloud edges with insufficient detail.

The proposed method achieves superior results due to its utilization of PVT blocks and CNN blocks as the feature extractor, enabling the model to comprehensively capture both global and local features from the remote sensing images. Additionally, the FAM is introduced to facilitate the effective integration of global and local features, enhancing the method's ability to grasp information at both scales. A BiDecoder is employed to reconstruct features from both top-down and bottom-up directions. To address the issue of rough segmentation boundaries, a boundary loss function is incorporated into the shallow layers of the feature extractor as an additional constraint. This allows the shallow layers to focus more on spatial detail, particularly object boundaries.

Experimental results show that the proposed method not only achieves more accurate segmentation for clouds of various scales compared to existing methods but also produces clearer segmentation boundaries with richer details.

2) Quantitative Comparison: Table V presents the quantitative comparison results. While the proposed method does not achieve the highest class accuracy scores (94.68% for clouds and 96.69% for the background), it is only marginally behind the top-performing methods. Moreover, several methods, including BiSeNetV1, DeepLabV3, UperNet, STDC, modified VGG, WiCoNet, GLNet, and MSLANet, show significant variance in accuracy across different categories. They may excel in one category but do not offer optimal comprehensive segmentation capabilities across various categories. In contrast, the proposed method demonstrates relatively small variance in accuracy across different categories and achieves the highest values in mAcc, mPrecision, mRecall, mF1score, and mIoU, with respective percentages of 95.94%, 95.64%, 95.68%, 95.66%, and 91.71%. This underscores the effectiveness of the proposed method on the dataset.

F. Ablation Study

Ablation studies are essential for evaluating and understanding the role and significance of each component within a model. They provide insight into the model's functionality and the contribution of each component to its overall performance. In this section, we perform ablation studies on the proposed method to thoroughly assess the effectiveness of the proposed modules, primarily using mIoU as the evaluation metric on the L8SPARCS dataset. We begin with PVT as the baseline and progressively incorporate the proposed modules to validate their effectiveness and the performance of the entire model. Table VI presents the results, showing that the model

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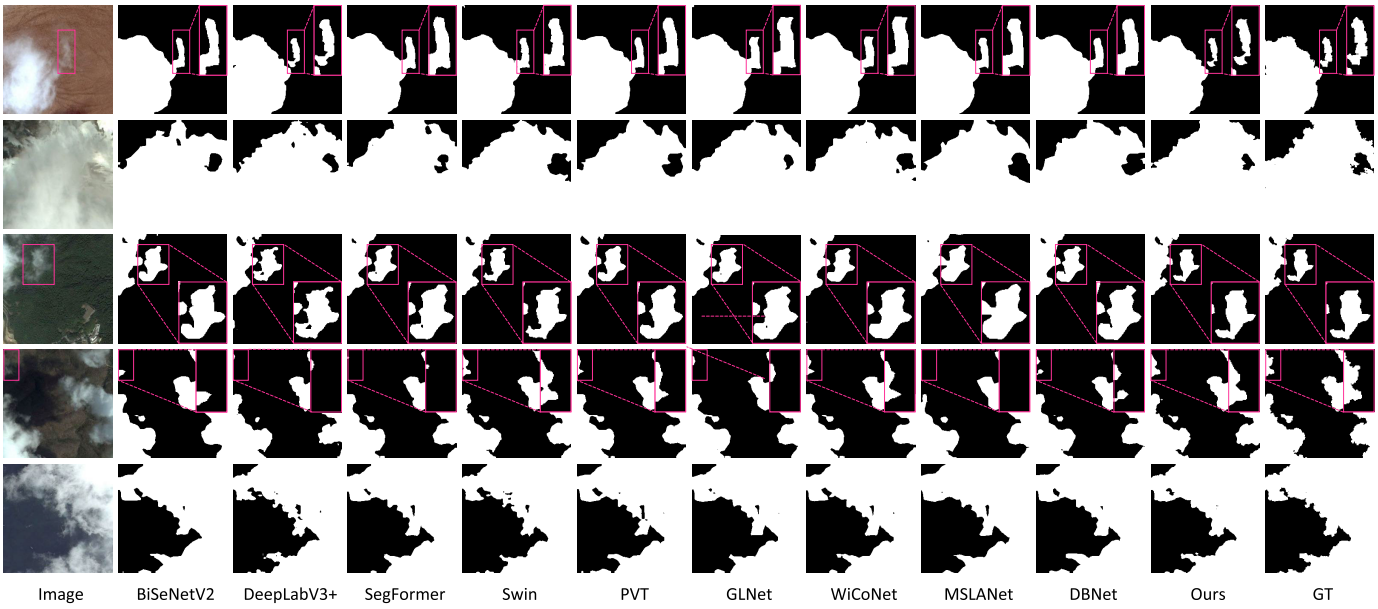


Fig. 7. Qualitative comparisons on the HRC_WHU dataset (white: cloud, black: background).

TABLE V
QUANTITATIVE COMPARISONS ON THE HRC_WHU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Class Accuracy		Overall Metrics				
	Cloud	Background	mAcc	mPrecision	mRecall	mF$_1$-score	mIoU
FCN8S [26]	93.56	95.43	94.73	94.27	94.49	94.38	89.39
DANet [47]	92.57	96.94	95.31	95.19	94.75	94.96	90.44
CCNet [48]	92.81	95.47	94.48	94.06	94.14	94.10	88.90
BiSeNetV1 [49]	91.27	96.56	94.59	94.47	93.91	94.18	89.03
BiSeNetV2 [50]	93.72	95.26	94.68	94.19	94.49	94.33	89.31
DeepLabV3 [30]	91.16	97.25	94.98	95.03	94.21	94.59	89.77
DeepLabV3+ [31]	92.03	97.00	95.15	95.07	94.52	94.78	90.11
HRNet [51]	93.04	96.54	95.23	95.00	94.79	94.89	90.31
FPN [52]	91.06	96.74	94.63	94.56	93.90	94.21	89.10
PSPNet [27]	93.12	97.32	95.75	95.67	95.22	95.44	91.30
OCRNet [53]	92.35	96.83	95.16	95.03	94.59	94.80	90.15
UPerNet [54]	91.63	**97.50**	95.31	95.38	94.57	94.95	90.41
NonLocal [55]	92.94	96.18	94.97	94.68	94.56	94.62	89.82
PSANet [56]	93.10	96.19	95.03	94.73	94.64	94.69	89.94
STDC [57]	91.07	97.00	94.79	94.79	94.04	94.39	89.41
ISANet [58]	93.95	96.75	95.71	95.46	95.35	95.41	91.24
SegFormer [59]	93.79	97.11	95.86	95.71	95.45	95.58	91.55
Swin [41]	**94.85**	95.75	95.12	94.94	95.30	95.12	90.71
modified VGG [32]	90.99	96.70	94.57	94.50	93.84	94.15	88.99
GAFFRNet [33]	94.35	95.53	95.09	94.61	94.94	94.77	90.09
MSLANet [34]	91.56	97.18	95.08	95.08	94.37	94.70	89.97
DBNet [39]	92.38	96.72	95.10	94.94	94.55	94.74	90.04
WiCoNet [43]	92.42	**97.50**	94.96	95.61	94.96	95.27	90.99
GLNet [44]	93.66	96.53	95.09	95.18	95.09	95.14	90.75
CTCFNet (ours)	94.68	96.69	**95.94**	**95.64**	**95.68**	**95.66**	**91.71**

incorporating all the proposed modules achieved the best performance.

1) PVT and CNN Blocks: The primary task of the feature extractor is to effectively extract high-quality feature representations from input data. Remote sensing images typically contain a variety of complex surface elements like buildings, roads, water bodies, vegetation, and clouds, which vary greatly in size, shape, and texture. Consequently, accurately and effectively extracting high-quality surface objects presents a challenge for the feature extractor. To address this, PVT

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TABLE VI
ABLATION FOR DIFFERENT MODULES IN THE PROPOSED METHOD

Methods	mIoU (%)
PVT (baseline)	81.92
PVT + CNN block	84.80 (↑**2.88**%)
PVT + CNN + FAM	85.24 (↑**0.44**%)
PVT + CNN + FAM + BiDecoder	85.73 (↑**0.49**%)
PVT + CNN + FAM + BiDecoder + Boundary loss	86.42 (↑**0.69**%)

TABLE VII
MIOU WITH DIFFERENT KERNELS IN BOUNDARY LOSS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Stage1	Stage2	Stage3	Stage4
Laplace	85.65	85.80	85.82	86.10
Prewitt	85.83	86.08	86.14	86.15
Roberts	85.82	**86.42**	85.83	85.63
Scharr	85.71	86.12	86.23	86.07
Sobel	85.91	85.57	85.93	85.88

blocks and CNN blocks are employed separately to capture the image features. Transformers and CNNs are two essential model architectures in computer vision, each offering distinct advantages: Transformers, with their self-attention mechanisms, excel at capturing global feature relationships. Their dynamic attention mechanism allows them to adaptively focus on different parts based on context, making them particularly effective in understanding complex, context-dependent high-resolution remote sensing images. Conversely, CNNs are adept at capturing local features due to their convolutional operations' translational invariance, which is well-suited for identifying local features in images, such as object boundaries and textures. The ablation study results indicate that incorporating the CNN block increases the mIoU of the baseline to 84.80% (↑2.88%). This substantial increase confirms the effectiveness of using PVT and CNN blocks together for feature extraction.

2) Feature Aggregation Module: The FAM is designed to combine the features extracted by the PVT blocks and the CNN blocks at different levels and of different types. This integration enhances the model's ability to understand complex scenes, enabling it to more effectively capture both local details and global semantic information. The FAM consists of three branches. The first branch, composed of strip poolings and a global pooling, further extracts edge and texture information at different scales based on the local features extracted by the CNN blocks. The second branch integrates the local features extracted by the CNN blocks with the global features extracted by the PVT blocks. The third branch adjusts the attention of features from the PVT blocks to different parts using the features integrated by the second branch. Finally, the features from these three branches are concatenated along the channel dimension and passed through a convolution to generate the final fused features. The inclusion of the FAM increases the mIoU from 84.80% to 85.24%, demonstrating its effectiveness for feature fusion.

3) Bidirectional Decoder: The primary function of the decoder is to reconstruct features from different levels to generate accurate pixel-level predictions. To achieve this, we employ a BiDecoder, which consists of two parallel subbranches: the top-down decoder and the bottom-up decoder. The top-down decoder starts with high-level abstract features from the deepest layer, progressively restoring spatial resolution and detail information. Conversely, the bottom-up decoder begins with features from the shallower layers, incorporating boundary and texture features into high-level abstract features to further enrich the spatial location information of the abstract semantic features. Incorporating the BiDecoder increases the mIoU of the proposed method to 85.73%, demonstrating its effectiveness in enhancing segmentation accuracy.

4) Boundary Loss: High-resolution remote sensing images often contain complex surface features and various types of objects, with boundaries that can be extremely subtle and difficult to distinguish. To address this, incorporating a boundary loss function into semantic segmentation methods for remote sensing images is necessary. The boundary loss function specifically targets these edge regions, enhancing the model's ability to learn and recognize boundary details. Accurately identifying the boundaries of various surface objects can significantly improve the model's precision and reliability. Table VI shows that after incorporating boundary loss function, the mIoU of the proposed method increases from 85.73% to 86.42%. The results, as indicated in Table VII, shows that using the Roberts kernel to generate boundary ground truth in the boundary loss function and applying this loss to Stage 2 of the feature extractor results in the highest mIoU, leading us to adopt the Roberts operator.

5) Attention Maps: Attention maps can highlight the areas or features that a model focuses on within an image, offering insights into how the model processes input data and makes predictions. However, it is crucial to understand that attention maps do not represent the final prediction results. While they reflect the distribution of the model's focus during input processing, they are not equivalent to the model's ultimate prediction outcomes.

Fig. 8 displays the attention maps for the proposed model on the L8SPARCS dataset as each component of the method is incrementally incorporated. It can be observed that PVT has a strong focus on global features but pays less attention to local details [Fig. 8(c)]. After incorporating the CNN blocks [Fig. 8(d)], the model's attention to local features, especially for medium- and small-scale objects, is significantly enhanced. Building on this, the introduced FAM effectively integrates the information extracted by PVT blocks and CNN blocks. By comparing the attention maps Fig. 8(e) with Fig. 8(c) and (d), it is evident that the proposed method with FAM not only captures global information better but also pays more attention to local details.

The proposed BiDecoder employs a parallel top-down and bottom-up approach, which better achieves multiscale information fusion and reduces information loss. Comparisons show that after incorporating the BiDecoder, the proposed method pays more attention to the detailed parts of objects at different scales.

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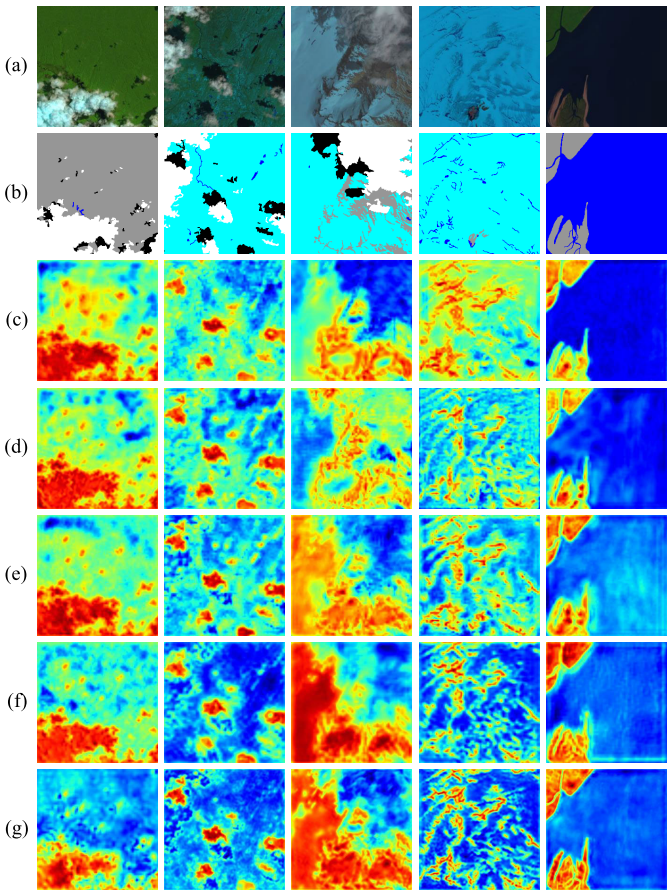


Fig. 8. Attention maps. (a) Images. (b) Ground truth. (c) PVT. (d) PVT + CNN blocks. (e) PVT + CNN blocks + FAM. (f) PVT + CNN blocks + FAM + BiDecoder. (g) PVT + CNN blocks + FAM + BiDecoder + boundary loss. From left to right, they are the attention maps of the proposed method for the categories cloud, cloud shadow, snow/ice, water, and land, respectively.

Edges are crucial for accurately segmenting object boundaries, enhancing overall accuracy, and leading to more refined segmentation results. To address this, a boundary loss function is added to the shallow layer of the feature extractor, encouraging the model to better distinguish between the boundaries of different objects and improving the clarity of the segmentation results. As seen in the attention maps [Fig. 8(g)], after adding the boundary loss, the model can more accurately capture the boundaries of objects and significantly reduce its attention to irrelevant surface objects.

V. Conclusion

This article proposes a CTCFNet for high-resolution remote sensing images semantic segmentation. CTCFNet consists of two main stages: feature extraction and feature fusion. In the feature extraction stage, a feature extractor comprising CNN and PVT blocks is utilized to capture both local and global complementary information. Additionally, a boundary loss function is proposed to enhance the method's ability to accurately segment textures and object boundaries in remote sensing images. Moving to the feature fusion stage, an FAM is employed to integrate global and local features. This integration allows these two types of information to complement each other effectively, thereby enhancing the representation of both local details and global context. Furthermore, a BiDecoder is implemented to reconstruct precise semantic segmentation results.

Through these design elements, CTCFNet achieves accurate semantic segmentation results for remote sensing images, particularly with enhanced boundary details. However, it should be noted that CTCFNet requires substantial computing resources, features a complex model architecture with a large number of parameters, and heavily relies on extensive high-quality labeled data. In future work, we plan to address these challenges by optimizing the model architecture, introducing adaptive mechanisms, enhancing data augmentation techniques, and developing lightweight designs to improve model efficiency without compromising performance.

References

[1] A. Li, L. Jiao, H. Zhu, L. Li, and F. Liu, "Multitask semantic boundary awareness network for remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2021.

[2] D. G. Manolakis, G. A. Shaw, and N. Keshava, "Comparative analysis of hyperspectral adaptive matched filter detectors," *Proc. SPIE*, vol. 4049, pp. 2–17, Aug. 2000.

[3] L. S. Bins, L. G. Fonseca, G. J. Erthal, and F. M. Ii, "Satellite imagery segmentation: A region growing approach," *Simpósio Brasileiro Sensoriamento Remoto*, vol. 8, no. 1996, pp. 677–680, 1996.

[4] A. W. Abbas, N. Minallh, N. Ahmad, S. A. R. Abid, and M. A. A. Khan, "K-means and isodata clustering algorithms for landcover classification using remote sensing," *Sindh Univ. Res. J.*, vol. 48, no. 2, pp. 315–318, 2016.

[5] X. Daxiang, T. Debao, W. Xiongfei, and W. Qiao, "A dynamic threshold cloud detecting approach based on the brightness temperature from FY-2 VISSR data," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 7, pp. 617–625, Apr. 2015.

[6] X. Liu, J.-M. Xu, and B. Du, "A bi-channel dynamic thershold algorithm used in automatically identifying clouds on GMS-5 imagery," *J. Appl. Meteorlog. Sci*, vol. 16, pp. 134–444, Jan. 2005.

[7] P. L. Rosin and J. Hervás, "Remote sensing image thresholding methods for determining landslide activity," *Int. J. Remote Sens.*, vol. 26, no. 6, pp. 1075–1092, Mar. 2005.

[8] A. V. Di Vittorio and W. J. Emery, "An automated, dynamic threshold cloud-masking algorithm for daytime AVHRR images over land," *IEEE Trans. Geosci. Remote Sens.*, vol. 40, no. 8, pp. 1682–1694, Aug. 2002.

[9] Z. Wang, J. R. Jensen, and J. Im, "An automatic region-based image segmentation algorithm for remote sensing applications," *Environ. Model. Softw.*, vol. 25, no. 10, pp. 1149–1165, Oct. 2010.

[10] G. M. Espindola, G. Camara, I. A. Reis, L. S. Bins, and A. M. Monteiro, "Parameter selection for region-growing image segmentation algorithms using spatial autocorrelation," *Int. J. Remote Sens.*, vol. 27, no. 14, pp. 3035–3040, Jul. 2006.

[11] T. Zhang, X. Yang, S. Hu, and F. Su, "Extraction of coastline in aquaculture coast from multispectral remote sensing images: Object-based region growing integrating edge detection," *Remote Sens.*, vol. 5, no. 9, pp. 4470–4487, Sep. 2013.

[12] Y. Gao, J. F. Mas, N. Kerle, and J. A. Navarrete Pacheco, "Optimal region growing segmentation and its effect on classification accuracy," *Int. J. Remote Sens.*, vol. 32, no. 13, pp. 3747–3763, Jul. 2011.

[13] B. Kaur and A. Garg, "Mathematical morphological edge detection for remote sensing images," in *Proc. 3rd Int. Conf. Electron. Comput. Technol.*, vol. 5, Apr. 2011, pp. 324–327.

[14] M. Ali and D. Clausi, "Using the Canny edge detector for feature extraction and enhancement of remote sensing images," in *Proc. Scanning Present Resolving Future, IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2001, pp. 2298–2300.

[15] S. Guiming and S. Jidong, "Remote sensing image edge-detection based on improved Canny operator," in *Proc. 8th IEEE Int. Conf. Commun. Softw. Netw. (ICCSN)*, Jun. 2016, pp. 652–656.

[16] X. Li, S. Zhang, X. Pan, P. Dale, and R. Cropp, "Straight road edge detection from high-resolution remote sensing images based on the ridgelet transform with the revised parallel-beam radon transform," *Int. J. Remote Sens.*, vol. 31, no. 19, pp. 5041–5059, Oct. 2010.

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[17] A. Ghosh, N. S. Mishra, and S. Ghosh, "Fuzzy clustering algorithms for unsupervised change detection in remote sensing images," *Inf. Sci.*, vol. 181, no. 4, pp. 699–715, Feb. 2011.
[18] H. Zhai, H. Zhang, L. Zhang, P. Li, and A. Plaza, "A new sparse subspace clustering algorithm for hyperspectral remote sensing imagery," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 43–47, Jan. 2017.
[19] Z. Lv, Y. Hu, H. Zhong, J. Wu, B. Li, and H. Zhao, "Parallel k-means clustering of remote sensing images based on mapreduce," in *Proc. Int. Conf. Web Inf. Syst. Mining*, Sanya, China. Cham, Switzerland: Springer, 2010, pp. 162–170.
[20] B. S. Chandana, K. Srinivas, and R. K. Kumar, "Clustering algorithm combined with Hill climbing for classification of remote sensing image," *Int. J. Electr. Comput. Eng. (IJECE)*, vol. 4, no. 6, p. 923, Dec. 2014.
[21] J. Yuan, D. Wang, and R. Li, "Remote sensing image segmentation by combining spectral and texture features," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 1, pp. 16–24, Jan. 2014.
[22] R. Trias-Sanz, G. Stamon, and J. Louchet, "Using colour, texture, and hierarchial segmentation for high-resolution remote sensing," *ISPRS J. Photogramm. Remote Sens.*, vol. 63, no. 2, pp. 156–168, Mar. 2008.
[23] R. Gaetano, G. Scarpa, and G. Poggi, "Hierarchical texture-based segmentation of multiresolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 7, pp. 2129–2141, Jul. 2009.
[24] A. Lucieer, A. Stein, and P. Fisher, "Multivariate texture-based segmentation of remotely sensed imagery for extraction of objects and their uncertainty," *Int. J. Remote Sens.*, vol. 26, no. 14, pp. 2917–2936, Jul. 2005.
[25] Y. Li and P. Gong, "An efficient texture image segmentation algorithm based on the GMRF model for classification of remotely sensed imagery," *Int. J. Remote Sens.*, vol. 26, no. 22, pp. 5149–5159, Nov. 2005.
[26] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.
[27] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2881–2890.
[28] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "Semantic image segmentation with deep convolutional nets and fully connected CRFs," 2014, *arXiv:1412.7062*.
[29] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.
[30] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*.
[31] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.
[32] Y. Zhan, J. Wang, J. Shi, G. Cheng, L. Yao, and W. Sun, "Distinguishing cloud and snow in satellite images via deep convolutional network," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 10, pp. 1785–1789, Oct. 2017.
[33] M. Xia, T. Wang, Y. Zhang, J. Liu, and Y. Xu, "Cloud/shadow segmentation based on global attention feature fusion residual network for remote sensing imagery," *Int. J. Remote Sens.*, vol. 42, no. 6, pp. 2022–2045, Mar. 2021.
[34] X. Dai, M. Xia, L. Weng, K. Hu, H. Lin, and M. Qian, "Multiscale location attention network for building and water segmentation of remote sensing image," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5609519.
[35] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 5998–6008.
[36] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.
[37] S. Zheng et al., "Rethinking semantic segmentation from a sequence-to-sequence perspective with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 6881–6890.
[38] W. Wang et al., "Pyramid vision transformer: A versatile backbone for dense prediction without convolutions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 568–578.
[39] C. Lu, M. Xia, M. Qian, and B. Chen, "Dual-branch network for cloud and cloud shadow segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5410012.
[40] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[41] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
[42] M. Jiang et al., "GraphGST: Graph generative structure-aware transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5504016.
[43] L. Ding et al., "Looking outside the window: Wide-context transformer for the semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410313.
[44] W. Chen, Z. Jiang, Z. Wang, K. Cui, and X. Qian, "Collaborative global-local networks for memory-efficient segmentation of ultra-high resolution images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 8916–8925.
[45] L. Zhang, Z. Tan, G. Zhang, W. Zhang, and Z. Li, "Learn more and learn usefully: Truncation compensation network for semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4403814.
[46] Q. Hou, L. Zhang, M.-M. Cheng, and J. Feng, "Strip pooling: Rethinking spatial pooling for scene parsing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4003–4012.
[47] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.
[48] Z. Huang, X. Wang, L. Huang, C. Huang, Y. Wei, and W. Liu, "CCNet: Criss-cross attention for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 603–612.
[49] C. Yu, J. Wang, C. Peng, C. Gao, G. Yu, and N. Sang, "BiseNet: Bilateral segmentation network for real-time semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 325–341.
[50] C. Yu, C. Gao, J. Wang, G. Yu, C. Shen, and N. Sang, "BiSeNet v2: Bilateral network with guided aggregation for real-time semantic segmentation," *Int. J. Comput. Vis.*, vol. 129, no. 11, pp. 3051–3068, Nov. 2021.
[51] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 5693–5703.
[52] A. Kirillov, R. Girshick, K. He, and P. Dollár, "Panoptic feature pyramid networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6399–6408.
[53] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2020, pp. 173–190.
[54] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 418–434.
[55] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.
[56] H. Zhao et al., "PSANet: Point-wise spatial attention network for scene parsing," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 267–283.
[57] M. Fan et al., "Rethinking BiSeNet for real-time semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 9716–9725.
[58] L. Huang, Y. Yuan, J. Guo, C. Zhang, X. Chen, and J. Wang, "Interlaced sparse self-attention for semantic segmentation," 2019, *arXiv:1907.12273*.
[59] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.



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